

Time Series

Decomposition: Seasonality Estimation

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Time Series Course 1GAMMA

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Course Objectives

By the end of this session, students will be able to:

- Understand the concept of seasonality in time series
- Estimate seasonal components using different methods
- Apply seasonal averaging
- Apply parametric regression for seasonality estimation
- Apply cosine-sine models
- Remove seasonality from time series data

Time Series Model

Time series decomposition:

$$Y_t = T_t + S_t + \varepsilon_t$$

Where:

- T_t : trend
- S_t : seasonality
- ε_t : stationary component

Goal:

- Estimate seasonality
- Remove seasonality

Seasonal Average Method

Seasonal component estimation:

$$\hat{S}_k = \frac{1}{n_k} \sum Y_t$$

where observations belong to season k

Deseasonalized series:

$$Y_t - \hat{S}_k$$

Example: Seasonal Average in R

```
# Monthly data example
month <- rep(1:12,10)
y <- 10 + sin(month/12*2*pi) + rnorm(120)

# Estimate seasonal means
season <- tapply(y, month, mean)

# Remove seasonality
y_deseason <- y - season[month]
```

Example: Seasonal Average in Python

```
import numpy as np
import pandas as pd

month = np.tile(np.arange(1,13),10)
y = 10 + np.sin(month/12*2*np.pi) + np.random.randn(120)

df = pd.DataFrame({"y":y,"month":month})

season = df.groupby("month").mean()

df["deseason"] = df["y"] - df["month"].map(season["y"])
```

Parametric Regression Method

Model using dummy variables:

$$Y_t = \beta_0 + \beta_1 D_1 + \cdots + \beta_d D_d + \varepsilon_t$$

Where dummy variables represent seasons.

Example: Seasonal Regression in R

```
month <- factor(month)

model <- lm(y ~ month)

season <- predict(model)
```

Example: Seasonal Regression in Python

```
import statsmodels.formula.api as smf

df["month"] = df["month"].astype("category")

model = smf.ols("y ~ C(month)", data=df).fit()

season = model.predict(df)
```

Cosine-Sine Model

Seasonality modeled as:

$$S_t = a \cos(\omega t) + b \sin(\omega t)$$

Where:

$$\omega = \frac{2\pi}{d}$$

Example: Cosine-Sine Model in R

```
omega <- 2*pi/12  
  
model <- lm(y ~ cos(omega*t) + sin(omega*t))  
  
season <- predict(model)
```

Example: Cosine-Sine Model in Python

```
omega = 2*np.pi/12

df["cos"] = np.cos(omega*t)
df["sin"] = np.sin(omega*t)

import statsmodels.api as sm

X = df[["cos","sin"]]
X = sm.add_constant(X)

model = sm.OLS(y,X).fit()

season = model.predict(X)
```

Deseasonalization

Remove seasonality:

$$Y_t^* = Y_t - \hat{S}_t$$

Result:

- Stationary series
- Ready for time series modeling

Summary

Methods covered:

- Seasonal averaging
- Regression with dummy variables
- Cosine-sine model